Quinn Gifford

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EDUCATION

Oregon State University — **Corvallis, OR**

B.S. in Electrical and Computer Engineering, Computer Science Minor | GPA: 3.70 — **June 2027 (Expected)**

TECHNICAL SKILLS

Hardware: Embedded Systems, PCBs, KiCad, Soldering, FPGA, Verilog, AVR, STM32, RTOS, Verification/Validation, PID

Software: C/C++/C#, Python, Matlab, Java, Javascript, TypeScript, SQL, Linux, React, Next.js, PyTorch, CI/CD, Claude

PROFESSIONAL EXPERIENCE

Systems Engineer Intern | Boeing — **June 2026 - Present**

- Contributing to verification, validation, and certification of the ERS (Earth Reference System) and ADIRS (Air Data and Inertial Reference System).

Software Engineer Intern | Alethian — **June 2025 - September 2025**

- Deployed a national-scale vector embedding database using Pinecone to store and index pharmacy data enriched with geolocation metadata and entry-type filters, enabling highly performant semantic queries.
- Built a high-speed pharmacy search API endpoint combining semantic vector search with geospatial proximity ranking; achieved **sub-100ms latency** for targeted queries and approximately 2 second response time for fallback geolocation queries in worst-case scenarios.
- Created performance testing tools for various applications such as insurance card scanning and speech recognition, allowing our team to make a **20% increase in model accuracy** through iterative improvements.

Software Engineer Intern | Scale — **May 2024 - September 2024**

- Pre-processed GitHub pull requests from SWE-bench subset for LLM training using GitHub API and Pandas, resulting in a **12% increase** in resolved programming problems during testing.
- Implemented RLHF training and testing procedures for tool use, API interactions, retrieval-augmented generation (RAG) with text extraction/web scraping, and Python-based mathematical libraries.
- Completed RLHF tasks, including writing programming related prompts, evaluating model generated code, and writing correct code for model training, in various languages such as Python, Java, JavaScript, Go, PHP, Verilog.

STEM Instructor | STEP IT Academy — **June 2023 - September 2023**

- Instructed classes of up to 10 students on various technology related subjects such as robotics, data structures and algorithms, game development, PC maintenance and repair, and more.
- Gave IT support by assisting with technology integration in the classrooms, troubleshooting technical issues, and maintaining digital learning resources.

PROJECTS (SEE WEBSITE FOR MORE→) — HTTPS://QUINNGIFFORDWEBSITE.VERCEL.APP/

Guitar Hero on FPGA

- Recreated Guitar Hero on a CMOD S7 **FPGA**, with a custom bare metal 4-button controller PCB and 4-LED strip display
- Designed and implemented the game logic using **VHDL/Verilog**, managing real-time button inputs and LED outputs

Munkey AI

- Built a full stack AI learning platform (TypeScript, Next.js, Supabase, Pinecone, OpenAI API) where teachers create courses, upload assignments and learning materials, manage grades, and teach students with the help of customizable AI.
- Automatically parses uploaded assignments and course materials to generate a dedicated and trackable AI chat for each problem for every student, powered by customizable system prompts and RAG for context-aware assistance.